REGULARISATION IN GUIDING

FRANK – CAMILLA (VU AMSTERDAM)

1. General setup

Consider discrete-time nonlinear Gaussian model

$$X_{t+1} \mid X_t = x \sim \mathcal{N}\left(b_t(x), Q_t(x)\right).$$

Assume to observe

$$V_t \mid X_t = x \sim K(x, \cdot),$$

Consider the continuous-time SDE model

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + q\,\mathrm{d}W_t, \qquad X_0 = x_0.$$

Assume to observe

$$V_i \mid X_{t_i} = x \sim K(x, \cdot).$$

We assume the Markov kernel K to admit a density $k(x, \cdot)$ with respect to Lebesgue measure.

Outline:

(1) focus on filtering using SMC
(2) Upon linear approximation of b and k, we can compute guiding functions
(3) Analyse behaviour of resulting guided process, show need of safeguarding using ε.

2. Discrete example setup

We consider the Markov kernel $P(x_0, dx_1)$ defined by

$$X_1 \mid X_0 \;\sim\; \mathcal{N}(\omega + \psi X_0,\; \eta^2).$$

We assume $v \sim x + \log Z^2$, where $Z \sim \mathcal{N}(0,1)$. Define

$$h(x_1) = \text{density of } X_1 + \log Z^2 \text{ evaluated at } v, \quad \text{where } Z \sim \mathcal{N}(0,1).$$

We have

$$P\big(X_1 + \log Z^2 \le v\big) = P\big(Z^2 \le e^{v - X_1}\big)$$

so

$$h(x_1) = f_{\chi^2(1)}\big(e^{v-x_1}\big)\, e^{v-x_1}.$$

For large x_1 it will behave like $\sqrt{e^{v-x_1}}$. It should also behave this way for large x_1. Define, using Gaussian approximation to the distribution of $\log Z^2$

$$g(x_1) = \varphi(v;\; \mu + x_1,\; \sigma^2), \qquad \mu = -1.27, \quad \sigma^2 = \pi^2/2,$$

where $\varphi(\,\cdot\,; \mu, \sigma^2)$ denotes the Gaussian density.

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We have twisted kernels

$$P^h(x_0, dx_1) = \frac{P(x_0, dx_1)\, h(x_1)}{\int P(x_0, dx_1)\, h(x_1)}, \qquad P^g(x_0, dx_1) = \frac{P(x_0, dx_1)\, g(x_1)}{\int P(x_0, dx_1)\, g(x_1)}.$$

Define $c_g = \int P(x_0, dx_1)\, g(x_1)$ and similarly c_h. As x_0 is fixed in this section, we omit it from our notation. In our specific case

$$c_g = \varphi(v; \mu + \omega + \psi x_0; \sigma^2 + \eta^2).$$

Moreover,

$$P^g(x_0,\, \mathrm{d}x_1) \sim \mathcal{N}\left(\mu^\star, (\sigma^\star)^2\right),$$

with

$$(\sigma^\star)^2 = \frac{\eta^2\sigma^2}{\eta^2 + \sigma^2} \qquad \mu^\star = \frac{\sigma^2(\omega + \psi x_0) + \eta^2(v - \mu)}{\eta^2 + \sigma^2}.$$

The measure obtained from g_ε is a mixture:

$$P^{g+\varepsilon}(x_0, dx_1) = \frac{g(x_1) + \varepsilon}{c_g + \varepsilon} P(x_0, dx_1) = \lambda_g(\varepsilon) P^g(x_0, dx_1) + (1 - \lambda_g(\varepsilon)) P(x_0, dx_1),$$

with $\lambda_g(\varepsilon) = c_g/(c_g + \varepsilon)$.

3. Sequential Monte Carlo Approach

Suppose we do sequential Monte Carlo to sample from $X_1 \mid v$ and to find the marginal likelihood $p(v)$, which is actually c_h. We use mutation kernel $M = P^{g+\varepsilon}$. Then it follows from the results in Chapter 5 of Chopin-Papaspiliopoulos that the potential should be

$$\begin{aligned} G_\varepsilon(x_1) &:= h(x_1)\, \frac{P(x_0, dx_1)}{P^{g+\varepsilon}(x_0, dx_1)} \\ &= h(x_1)\, \frac{\int P(x_0, dx_1)\, g(x_1)\, dx_1 + \varepsilon}{g(x_1) + \varepsilon} \\ &= h(x_1)\, \frac{c_g + \varepsilon}{g(x_1) + \varepsilon}. \end{aligned}$$

The estimator for the marginal likelihood is

$$\hat{L} := \frac{1}{N} \sum_{n=1}^{N} G_\varepsilon\left(X_1^{(n)}\right), \qquad (X_1^{(n)})_{n=1}^N \sim P^{g+\varepsilon}(x_0, dx_1).$$

For *any* $\varepsilon > 0$

$$\mathbb{E}_{P^{g+\varepsilon}} \hat{L} = \mathbb{E}_{P^{g+\varepsilon}} G_\varepsilon(X_1) = \int h(x_1)\, \frac{P(x_0, dx_1)}{P^{g+\varepsilon}(x_0, dx_1)} P^{g+\varepsilon}(x_0, dx_1) = c_h.$$

3.1. Analysis of variance of $\hat{L}$.

Define

$$m(\varepsilon) = \mathbb{E}_{P^{g+\varepsilon}} \left[G_\varepsilon(x_0, X_1)\right]^2.$$

By expanding we get

$$\begin{aligned} m(\varepsilon) &= \int h(x_1)^2 \left(\frac{c_g + \varepsilon}{g(x_1) + \varepsilon}\right)^2 \frac{P(x_0, dx_1)\,(g(x_1) + \varepsilon)}{c_g + \varepsilon} \\ &= \left(c_g + \varepsilon\right) \int \frac{h(x_1)^2}{g(x_1) + \varepsilon}\, P(x_0, dx_1). \end{aligned} \tag{1}$$

Define

$$I(\varepsilon) = \int \frac{h(x_1)^2}{g(x_1)+\varepsilon}\, P(x_0, dx_1)$$

and

$$\bar{m}(\varepsilon) = \log m(\varepsilon) = \log(c_g + \varepsilon) + \log I(\varepsilon).$$

Note that we can rewrite to

$$I(\varepsilon) = \int h(x_1)\, \frac{\int h(x_1)\, P(x_0, dx_1)}{g(x_1)+\varepsilon}\, P^h(x_0, dx_1) = c_h \int \frac{h(x_1)}{g(x_1)+\varepsilon}\, P^h(x_0, dx_1).$$

though we won't use this expression in the numerics.

3.2. **Infinite variance case.** Suppose $\varepsilon = 0$. Since $c > 0$, finiteness of $\bar{m}(0)$ is equivalent to finiteness of

$$I(0) = \int \frac{h(x_1)^2}{g(x_1)}\, P(x_0, dx_1).$$

Using $f_{\chi^2(1)}(u) = (2\pi u)^{-1/2} e^{-u/2}$ we get

$$h(x_1) = \frac{1}{\sqrt{2\pi}}\, e^{-(v-x_1)/2}\, e^{-e^{v-x_1}/2},$$

so

$$h(x_1)^2 = \frac{1}{2\pi}\, e^{-(v-x_1)}\, e^{-e^{v-x_1}}.$$

Writing $t = x_1 - v$, the integrand of $I(0)$ equals

$$\frac{h(x_1)^2}{g(x_1)}\, p(x_1 \mid x_0) \;\propto\; \exp\!\left(t - e^{-t} + \frac{(t+\mu)^2}{2\sigma^2} - \frac{(t - (\omega + \psi x_0 - v))^2}{2\eta^2} \right).$$

We analyse the tails:

- *Right tail* $t \to +\infty$. The term $e^{-t} \to 0$, and expanding the quadratics the dominant behaviour is

  $$\exp\!\left(t^2 \left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2} \right) + O(t) \right).$$

  This is integrable if and only if the coefficient of t^2 is strictly negative, i.e.

  $$\frac{1}{2\sigma^2} - \frac{1}{2\eta^2} < 0 \iff \eta^2 < \sigma^2.$$

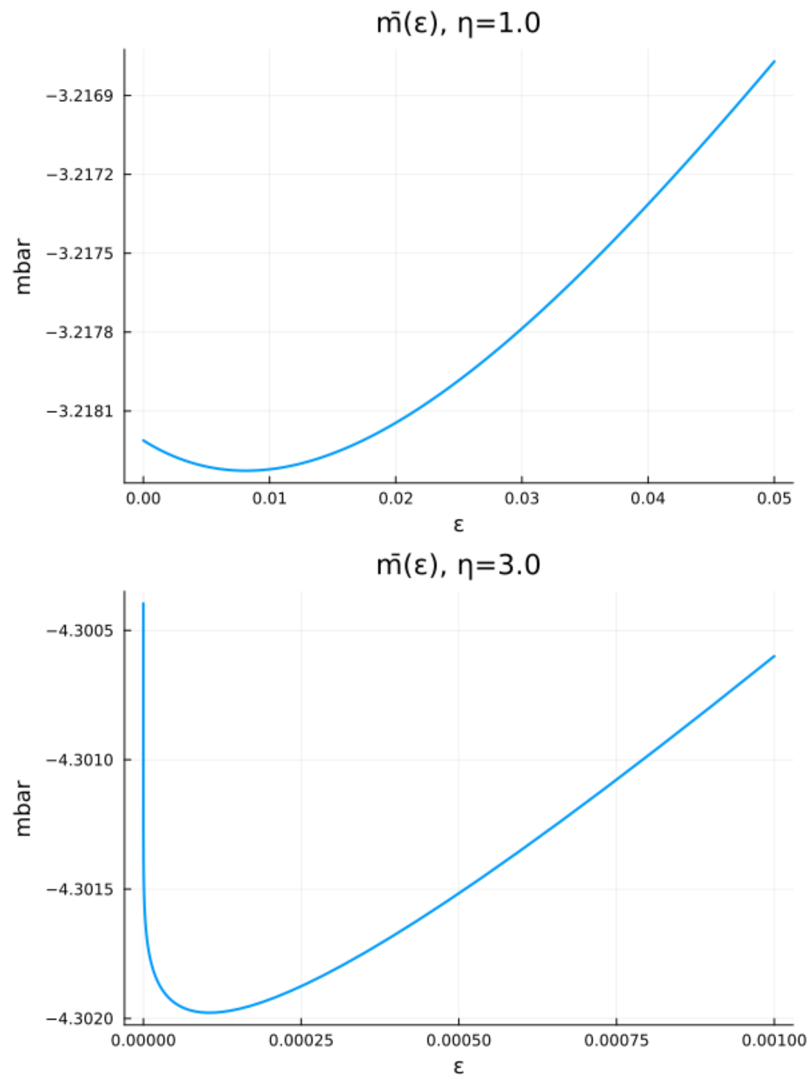
- *Left tail* $t \to -\infty$. The term $e^{-t} \to +\infty$ dominates inside the exponent, sending the integrand to zero faster than any polynomial. Hence the left tail never causes divergence.

Therefore,

Proposition 1. With the observation scheme $v = X_1 + \log Z^2$, $Z \sim \mathcal{N}(0,1)$, we have

$$I(0) = \int \frac{h(x_1)^2}{g(x_1)}\, P(x_0, dx_1) < \infty \iff \eta < \sigma.$$

It follows that for $\eta < \pi/\sqrt{2} \approx 2.221$ no regularisation is needed. For $\eta = \sigma$ the integral diverges (though like e^t), while for $\eta > \sigma$ divergence is much faster.


m̄(ε), η=1.0
mbar
−3.2169
−3.2172
−3.2175
−3.2178
−3.2181
0.00
0.01
0.02
0.03
0.04
0.05
ε
m̄(ε), η=3.0
mbar
−4.3005
−4.3010
−4.3015
−4.3020
0.00000
0.00025
0.00050
0.00075
0.00100
ε

4. Importance Sampling Quality: KL and χ^2 Divergences

4.1. **General theory.** The natural divergences for assessing IS quality of $\hat{L}$ are the KL divergence and the χ^2 divergence between the target P^h and the proposal $P^{g+\varepsilon}$:

$$\mathrm{KL}(P^h \| P^{g+\varepsilon}) = \int \log \frac{dP^h}{dP^{g+\varepsilon}}\, dP^h,$$

$$\chi^2(P^h \| P^{g+\varepsilon}) = \int \left(\frac{dP^h}{dP^{g+\varepsilon}}\right)^2 dP^{g+\varepsilon} - 1.$$

Since

$$\frac{dP^h}{dP^{g+\varepsilon}}(x_1) = \frac{h(x_1)/c_h}{(g(x_1)+\varepsilon)/(c_g+\varepsilon)} = \frac{G_\varepsilon(x_1)}{c_h},$$

we have

$$\chi^2(P^h \| P^{g+\varepsilon}) = \frac{m(\varepsilon)}{c_h^2} - 1,$$

so finite variance of the IS estimator is *exactly equivalent* to $\chi^2(P^h \| P^{g+\varepsilon}) < \infty$. The χ^2 divergence thus directly controls the variance. Note, using $\log x \le x - 1$ (and thus $2\log x \le x^2 - 1$)

$$2\mathrm{KL}(P^h \| P^{g+\varepsilon}) \le \int \left(\frac{dP^h}{dP^{g+\varepsilon}}\right)^2 dP^h - 1.$$

Original bound I wrote down was wrong.

4.2. **KL divergence: explicit form.** We have

$$\mathrm{KL}(P^h \| P^{g+\varepsilon}) = \log \frac{c_g+\varepsilon}{c_h} + \int \log h(x_1)\, P^h(x_0, dx_1) - \int \log(g(x_1)+\varepsilon)\, P^h(x_0, dx_1).$$

As $\varepsilon \downarrow 0$ each term has a finite limit provided $\int |\log g(x_1)|\, P^h(x_0, dx_1) < \infty$. Since $\log g(x_1)$ is quadratic in x_1 (as g is Gaussian), this requires P^h to have finite second moments. Hence

$$\mathrm{KL}(P^h \| P^{g+\varepsilon}) \to \mathrm{KL}(P^h \| P^g) < \infty \quad \text{as } \varepsilon \downarrow 0,$$

whenever P^h has finite second moments, which holds for any $\varepsilon \ge 0$ in our specific example.

4.3. **χ^2 divergence and finite variance: the threshold.** For any $\varepsilon > 0$, the denominator $g(x_1) + \varepsilon \ge \varepsilon > 0$ uniformly, so

$$m(\varepsilon) \le \varepsilon^{-1} \int h(x_1)^2\, P(x_0, dx_1) < \infty,$$

provided $h \in L^2(P(x_0, \cdot))$, which holds for our specific h. Hence the IS estimator has finite variance for *every* $\varepsilon > 0$.

At $\varepsilon = 0$ the situation is more delicate. From the tail analysis of Section 2, the integrand $h(x_1)^2/g(x_1)$ behaves like

$$\exp\left(t^2\left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2}\right) + O(t)\right) \quad \text{as } t = x_1 - v \to +\infty,$$

leading to the following result.

Proposition 2. For the observation scheme $v = X_1 + \log Z^2$, $Z \sim \mathcal{N}(0,1)$:

$$\chi^2(P^h \| P^g) < \infty \iff m(0) < \infty \iff \eta < \sigma.$$

Now

$$ESS \approx \frac{N}{1 + \chi^2(P^h \| P^{g+\varepsilon})}.$$

So minimising $m(\varepsilon)$ at each SMC step is equivalent to maximising ESS.

5. Numerical Experiment: Effect of ε on SMC Performance

5.1. **Setup.** We consider the hidden Markov model

$$X_t \mid X_{t-1} \sim \mathcal{N}(\omega + \psi X_{t-1},\, \eta^2),$$
$$V_t \mid X_t = X_t + \log Z_t^2, \quad Z_t \sim \mathcal{N}(0,1),$$

with parameters $\omega = 0$, $\psi = 0.9$, $\mu = -1.27$, $\sigma^2 = \pi^2/2$. We compare two regimes:

- **Regime A** ($\eta = 1.0 < \sigma \approx 2.22$): finite-variance regime, where $m(0) < \infty$.
- **Regime B** ($\eta = 3.0 > \sigma$): infinite-variance regime, where $m(0) = \infty$.

In both regimes we run SMC with the twisted proposal $P^{g+\varepsilon}$ for several values of

$$\varepsilon \in \{0, 0.001, 0.01, 0.05, 0.1\}.$$

5.2. **SMC Algorithm.** At each time step t, the twisted proposal is the mixture

$$P^{g+\varepsilon}(x,\cdot) = \frac{c(x)}{c(x)+\varepsilon}\, P^g(x,\cdot) + \frac{\varepsilon}{c(x)+\varepsilon}\, P(x,\cdot),$$

where both $c(x)$ and the parameters of P^g are available in closed form (see Section 2). The IS weight for a particle x_t proposed from $P^{g+\varepsilon}(x_{t-1},\cdot)$ is

$$G_\varepsilon(x_{t-1}, x_t) = \frac{h_t(x_t)}{g_t(x_t)+\varepsilon}\,(c(x_{t-1}) + \varepsilon).$$

To assess performance, we run Algorithm 1 independently $R = 500$ times on the same simulated dataset ($T = 50$, $N = 2000$) and record both the minimum ESS over time and the maximum normalised weight over time for each run. Define

$$\mathrm{ESS}_{\min} = \min_{t=1,\dots,T} \mathrm{ESS}_t,$$
$$Q_{\max} = \max_{t=1,\dots,T} \max_n W_t^n.$$

A low $\mathrm{ESS}_{\min}$ indicates that at some point the particle cloud collapsed to a near-degenerate distribution, with a single particle carrying almost all the weight. similarly a large $Q_{\max}$ value indicates this.

5.3. **Results.** Figure 1 shows the empirical CDF of $\mathrm{ESS}_{\min}$ across $R = 500$ runs for each value of ε, separately for the two regimes. We took $N = 500$ particles for a dataset with $T = 50$.

- *Regime A ($\eta < \sigma$).* One can see an improvement upon taking $\epsilon > 0$, though $\epsilon = 0.01$ is clearly too large. One can see that the minimal ESS is about 60 for $\epsilon = 0$. This confirms the theoretical prediction: when $m(0) < \infty$, the IS weights have finite variance. While finiteness is guaranteed, it is typically beneficial to take nonzero ε. The value $m(0)$ can still be large, while finite.

Algorithm 1 Guided particle filter (GPF) with twisted proposal $P^{g+\varepsilon}$, following **?**, Algorithm 10.5

Require: observations $v_{1:T}$, particle count N, perturbation $\varepsilon \geq 0$, ESS threshold $\mathrm{ESS}_{\min}$

For $n = 1, \dots, N$: draw $X_0^n \sim P(x_0)$, set $w_0^n \leftarrow G_0(X_0^n)$, $W_0^n \leftarrow w_0^n / \sum_{m=1}^N w_0^m$ where $G_0(x_0) := P_0(dx_0)\, h_0(x_0)/M_0(x_0)$

for $t = 1, \dots, T$ **do**

  if $\mathrm{ESS}(W_{t-1}^{1:N}) < \mathrm{ESS}_{\min}$ **then**

    $A_{t-1}^{1:N} \leftarrow \mathrm{resample}(W_{t-1}^{1:N})$

    $\hat{w}_{t-1} \leftarrow 1$

  else

    $A_{t-1}^n \leftarrow n$

    $\hat{w}_{t-1} \leftarrow w_{t-1}^n$

  end if

  For $n = 1, \dots, N$: draw $X_t^n \sim M_t(X_{t-1}^{A_{t-1}^n}, dx_t) = P^{g_t+\varepsilon}(X_{t-1}^{A_{t-1}^n}, dx_t)$ by sampling from $P^{g_t}(X_{t-1}^{A_{t-1}^n}, \cdot) = \mathcal{N}(\mu^*, \sigma^{*2})$ with probability $\alpha^n = c(X_{t-1}^{A_{t-1}^n})/(c(X_{t-1}^{A_{t-1}^n}) + \varepsilon)$, and from $P(X_{t-1}^{A_{t-1}^n}, \cdot)$ otherwise

  $w_t^n \leftarrow \hat{w}_{t-1}^n \cdot G_t(X_{t-1}^{A_{t-1}^n}, X_t^n)$ where

$$G_t(x_{t-1}, x_t) := \frac{h_t(x_t)}{g_t(x_t) + \varepsilon} \cdot \big(c(x_{t-1}) + \varepsilon\big)$$

  $W_t^n \leftarrow w_t^n \,/\, \sum_{m=1}^N w_t^m$

end for

return $\sum_{t=1}^T \log \ell_t^N, \quad \{\mathrm{ESS}(W_t^{1:N})\}_{t=1}^T$ where, following equation (10.3) of **?**,

$$\ell_t^N := \begin{cases} \dfrac{1}{N}\displaystyle\sum_{n=1}^N w_t^n & \text{if resampling occurred at time } t, \\ \Big(\sum_{n=1}^N w_t^n\Big) \,/\, \Big(\sum_{n=1}^N w_{t-1}^n\Big) & \text{otherwise.} \end{cases}$$

- *Regime B ($\eta > \sigma$).* The picture changes dramatically. At $\varepsilon = 0$ the CDF has substantial mass near zero, meaning that in a significant fraction of runs the particle cloud collapses to near-degeneracy at some time step. Taking $\varepsilon = 0.01$ shifts the CDF markedly rightward, essentially eliminating the collapses. This is the direct numerical manifestation of $m(0) = \infty$: the IS weights have infinite variance, leading to occasional but catastrophic weight concentration on a single particle.

5.4. **On the difficulty of choosing ε in Regime B.** The results highlight that the choice of ε in Regime B is delicate. The optimal ε^* exists and can in principle be found by minimising $m(\varepsilon)$ numerically, as described in Section 3. However, $m(\varepsilon)$ depends on x_0 and varies across time steps as the particle cloud evolves, so a single fixed ε cannot be globally optimal. Furthermore, the performance is sensitive to this choice: the 5th percentile of $\mathrm{ESS}_{\min}$ drops sharply on both sides of ε^*, leaving a narrow window of well-performing values.

This suggests that a fixed ε is fundamentally limited in Regime B, and that an *adaptive* scheme — choosing ε_t at each step based on the current particle cloud by minimising an empirical estimate of $m(\varepsilon)$ — would be more robust. In Regime A, by contrast, $\varepsilon = 0$ is already satisfactory.

5.5. **Work by Chatterjee & Diaconis.** Suggest to look at $Q_{\max}$-statistic. See Figure 2. is maximum of the normalised weights. Note that $Q_{\max} \in [0, 1]$ and we want it to be small.

6. Adaptive Choice of ε_t^*

The following result motivates the adaptive estimator. For a fixed particle x_{t-1} and observation v_t, under $X_t \sim P^{g_t+\varepsilon}(x_{t-1}, \cdot)$, the second moment of the IS weight equals

$$m(\varepsilon, x_{t-1}) = \big(c(x_{t-1}) + \varepsilon\big) \int \frac{h_t(x_1)^2}{g_t(x_1) + \varepsilon} P(x_{t-1}, dx_1).$$

Given the particle cloud $\{X_{t-1}^n\}$ with normalised weights $\{W_{t-1}^n\}$, the population-level second moment is the weighted average

$$\tilde{m}(\varepsilon) = \sum_{n=1}^{N} W_{t-1}^n \cdot m(\varepsilon, X_{t-1}^n) = \sum_{n=1}^{N} W_{t-1}^n \big(c(X_{t-1}^n) + \varepsilon\big) \int \frac{h_t(x_1)^2}{g_t(x_1) + \varepsilon} P(X_{t-1}^n, dx_1).$$

So we evaluate $m(\varepsilon)$ by taking the expectation of x_{t-1} over the current particle approximation. The idea is to choose

$$\varepsilon_t^* = \underset{\varepsilon \geq 0}{\operatorname{argmin}}\ \tilde{m}(\varepsilon).$$

The integral appearing in $\tilde{m}(\varepsilon)$ can be approximately using numerical quadrature, but in our implementation we approximate it by drawing $\tilde{X}_t^n \sim P(X_{t-1}^n, \cdot)$ once and computing

$$\hat{m}(\varepsilon) = \sum_{n=1}^{N} W_{t-1}^n \frac{h_t(\tilde{X}_t^n)^2}{g_t(\tilde{X}_t^n) + \varepsilon} \cdot \big(c(X_{t-1}^n) + \varepsilon\big).$$

Thus, this is a Monte-Carlo estimator for $\tilde{m}(\varepsilon)$ based on N samples. The analytical derivative with respect to ε is

$$\frac{d\hat{m}}{d\varepsilon} = \sum_{n=1}^{N} W_{t-1}^n \, h_t(\tilde{X}_t^n)^2 \, \frac{g_t(\tilde{X}_t^n) - c(X_{t-1}^n)}{(g_t(\tilde{X}_t^n) + \varepsilon)^2},$$

which enables gradient-based optimisation. An alternative consists of minimising $\hat{m}(\varepsilon)$ over a log-spaced grid $\mathcal{E} = \{\varepsilon_1 < \cdots < \varepsilon_K\}$.

Remarks.

- The auxiliary proposals $\{\tilde{X}_t^n\}$ are drawn from the untwisted kernel P and discarded after the grid search. Since they do not enter the weight update, the likelihood estimator $\sum_t \log \ell_t^N$ remains unbiased for $\log p(v_{1:T})$ for any data-dependent choice of ε_t^*.

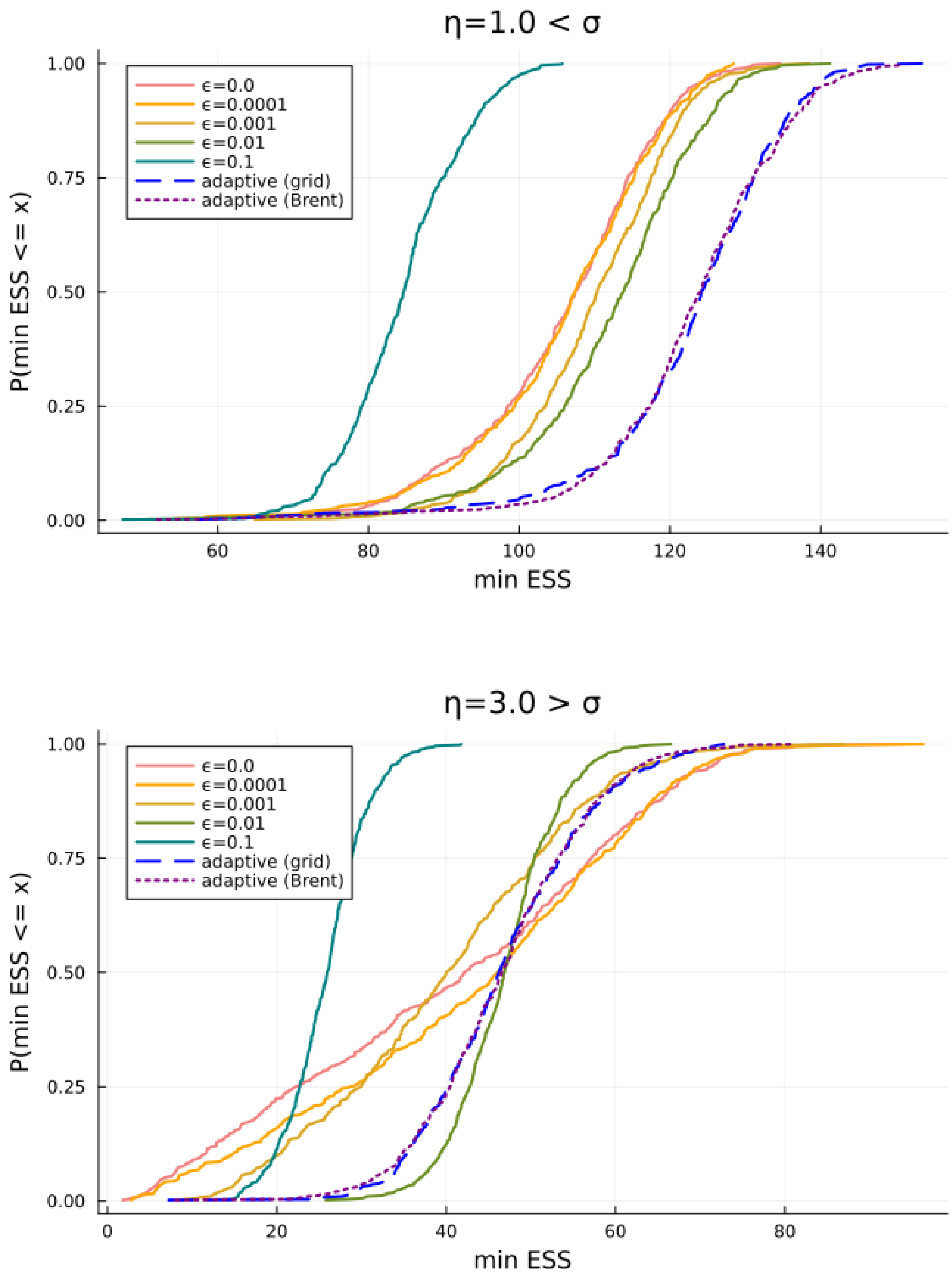


Figure 1. minESS comparison.

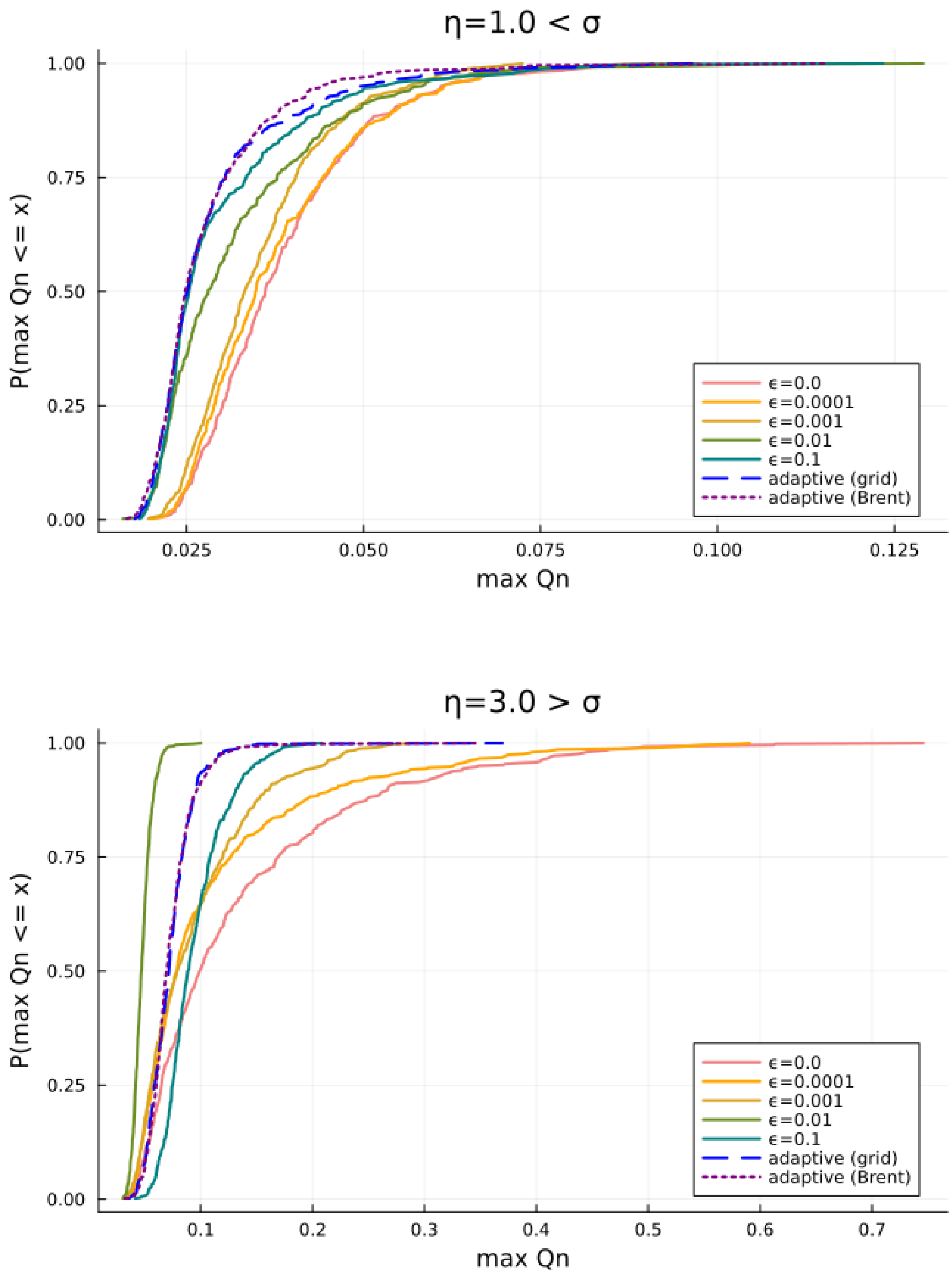


FIGURE 2. Max Q_n comparison.

Algorithm 2 Adaptive GPF with ε_t^* chosen by weighted grid search

Require: observations $v_{1:T}$, particle count N, ESS threshold $\mathrm{ESS}_{\min}$, grid $\mathcal{E} = \{\varepsilon_1 < \cdots < \varepsilon_K\}$

For $n = 1, \ldots, N$: draw $X_0^n \sim \mathcal{N}(0, \eta^2/(1-\psi^2))$, set $W_0^n \leftarrow 1/N$, $w_0^n \leftarrow 1$

for $t = 1, \ldots, T$ **do**

  Find ε_t^***:** draw auxiliary proposals $\tilde{X}_t^n \sim P(X_{t-1}^n, \cdot\,)$ for $n = 1, \ldots, N$, and set

$$\varepsilon_t^* = \operatorname*{argmin}_{\varepsilon \in \mathcal{E}} \ \hat{m}(\varepsilon), \qquad \hat{m}(\varepsilon) = \sum_{n=1}^{N} W_{t-1}^n \, \frac{h_t(\tilde{X}_t^n)^2}{g_t(\tilde{X}_t^n) + \varepsilon} \cdot \big(c(X_{t-1}^n) + \varepsilon\big)$$

  Discard $\{\tilde{X}_t^n\}$ after this step.

  if $\mathrm{ESS}(W_{t-1}^{1:N}) < \mathrm{ESS}_{\min}$ **then**

    $A_{t-1}^{1:N} \leftarrow \mathrm{resample}(W_{t-1}^{1:N}), \quad \hat{w}_{t-1}^n \leftarrow 1$

  else

    $A_{t-1}^n \leftarrow n, \quad \hat{w}_{t-1}^n \leftarrow w_{t-1}^n$

  end if

  Propose: for $n = 1, \ldots, N$ draw $X_t^n \sim P^{g_t + \varepsilon_t^*}\Big(X_{t-1}^{A_{t-1}^n}, \cdot\,\Big)$ by sampling from $P^{g_t}(X_{t-1}^{A_{t-1}^n}, \cdot) = \mathcal{N}(\mu^*, \sigma^{*2})$ with probability $\alpha^n = c(X_{t-1}^{A_{t-1}^n})/(c(X_{t-1}^{A_{t-1}^n}) + \varepsilon_t^*)$, and from $P(X_{t-1}^{A_{t-1}^n}, \cdot)$ otherwise

  Update weights:

$$w_t^n = \hat{w}_{t-1}^n \cdot G_t\Big(X_{t-1}^{A_{t-1}^n},\, X_t^n\Big), \qquad G_t(x_{t-1}, x_t) := \frac{h_t(x_t)}{g_t(x_t) + \varepsilon_t^*} \cdot \big(c(x_{t-1}) + \varepsilon_t^*\big)$$

  Likelihood increment (eq. 10.3 of **?**):

$$\ell_t^N := \begin{cases} \dfrac{1}{N} \displaystyle\sum_{n=1}^{N} w_t^n & \text{if resampling occurred at time } t, \\ \Big(\sum_{n=1}^{N} w_t^n\Big) \,/\, \Big(\sum_{n=1}^{N} w_{t-1}^n\Big) & \text{otherwise} \end{cases}$$

  Normalise: $W_t^n \leftarrow w_t^n \,/\, \sum_{m=1}^{N} w_t^m$

  ESS: $\mathrm{ESS}_t \leftarrow \big(\sum_n (W_t^n)^2\big)^{-1}$

end for

return $\sum_{t=1}^{T} \log \ell_t^N, \quad \{\mathrm{ESS}_t\}_{t=1}^{T}, \quad \{\varepsilon_t^*\}_{t=1}^{T}$

- Using the carry-over weights W_{t-1}^n in $\hat{m}(\varepsilon)$ is essential when resampling has not occurred recently. After resampling all weights equal $1/N$ and the weighted and unweighted averages coincide; in steps without resampling the weight distribution can be skewed and the unweighted average gives a poorly-targeted ε_t^*. Numerical experiments confirm that the weighted version gives a substantial improvement in both min-ESS and $Q_{\max}$ diagnostics.
- The grid $\mathcal{E}$ is log-spaced over $[10^{-6}, 1]$ with $K = 50$ points. All K evaluations of $\hat{m}(\varepsilon)$ reuse the same proposals $\{\tilde{X}_t^n\}$, so the cost is $O(KN)$ per step. Reducing to $K = 20$ or $K = 30$ would roughly halve the cost with negligible impact on ε_t^*.

7. Adaptive Choice of ε via ESS Targeting

This does not work; I guess the desired ESS simply cannot be attained.

7.1. **Motivation.** The numerical experiments of Section 4 show that the optimal perturbation ε^* varies across time steps and depends on the current particle cloud. A fixed ε cannot be globally optimal. We propose an adaptive scheme inspired by tempering methods in SMC **??**, where a tempering parameter is chosen at each step so that the ESS after reweighting equals a target fraction ρN. The same principle applies here: we choose ε_t^* such that the ESS of the IS weights equals ρN.

7.2. **ESS as a Function of ε.** Given proposals $\{x_t^{(i)}\}_{i=1}^N$ drawn from $P(x_{t-1}^{(i)}, \cdot)$, the ESS of the weights $w^{(i)}(\varepsilon) = h_t(x_t^{(i)})/(g_t(x_t^{(i)}) + \varepsilon)$ is

$$\mathrm{ESS}(\varepsilon) = \frac{\left(\sum_i w^{(i)}(\varepsilon)\right)^2}{\sum_i w^{(i)}(\varepsilon)^2}.$$

Two key properties make this suitable for bisection:

(1) $\mathrm{ESS}(\varepsilon)$ is *increasing* in ε: as $\varepsilon \to \infty$, all weights converge to $h_t(x_t^{(i)})/\varepsilon$ and become equal, so $\mathrm{ESS} \to N$.
(2) $\mathrm{ESS}(0)$ may be small or even zero if some $g_t(x_t^{(i)}) = 0$, reflecting the weight degeneracy in the infinite-variance regime.

The target ε_t^* is therefore the unique solution of

$$\mathrm{ESS}(\varepsilon_t^*) = \rho N,$$

found by bisection on $\varepsilon \in [0, \varepsilon_{\max}]$. If $\mathrm{ESS}(0) \geq \rho N$ already, we set $\varepsilon_t^* = 0$; if even $\varepsilon_{\max}$ is insufficient, we set $\varepsilon_t^* = \varepsilon_{\max}$.

7.3. **Connection to $m(\varepsilon)$.** The ESS targeting criterion is closely related to minimising $m(\varepsilon)$. Note that

$$\frac{\mathrm{ESS}(\varepsilon)}{N} \approx \frac{1}{1 + \chi^2(P^h \| P^{g+\varepsilon})} = \frac{c_h(x_0)^2}{m(\varepsilon)},$$

so targeting $\mathrm{ESS}(\varepsilon) = \rho N$ is equivalent to targeting $\chi^2(P^h \| P^{g+\varepsilon}) = (1-\rho)/\rho$, i.e. a fixed level of the χ^2 divergence. Rather than minimising $m(\varepsilon)$ (which requires a grid search and gives a different target at each step), ESS targeting gives a directly interpretable, step-by-step criterion for particle diversity, and bisection finds the solution efficiently without a grid.

8. Conclusion

Figure 3 shows the results of a single SMC run for both regimes and both fixed ($\varepsilon = 0$) and adaptive ε^*. Even for $\eta = 3$ the results with $\varepsilon = 0$ can be just fine: a single run looks almost indistinguishable from the adaptive run in terms of ESS, cumulative log-likelihood, and incremental log-likelihood. However, if one were to repeat the analysis many times, occasionally this would result in a catastrophic drop in ESS. This is precisely the implication of $m(0) = \infty$: it does not mean every run fails, but that the variance of the IS weights across runs is infinite, so disasters occur with positive probability and cannot be ruled out from a single run alone. This is confirmed by the CDF plots of

Figure 1, which required $R = 200$ independent runs to reveal the heavy left tail of the min-ESS distribution at $\varepsilon = 0$.

Two further observations support this interpretation. First, the incremental log-likelihood panel shows a pronounced dip around $t = 30$ in both the fixed and adaptive runs for $\eta = 3$. This dip is driven by a particular difficult observation and occurs regardless of ε, suggesting it reflects a genuinely hard step in the data rather than an algorithmic failure. In a catastrophic run, this dip would be far deeper and the ESS would collapse to near 1 at that point. Second, the adaptive ε^* panel for $\eta = 3$ shows ε^* spiking upward precisely around $t = 30$, at the same difficult observation. This confirms that the adaptive scheme is responding correctly: it detects the hard step through the empirical minimiser of $\hat{m}(\varepsilon)$ and automatically increases ε to protect against weight collapse, without any explicit knowledge of which observations are difficult.

9. Laplace Observation Scheme

9.1. **Setup.** As an alternative to the log-chi-squared observation scheme, consider

$$V_t = X_t + U_t, \qquad U_t \sim \text{Laplace}(0, b),$$

so that the observation density evaluated at v is

$$h(x_1) = \frac{1}{2b} \exp\left(-\frac{|v - x_1|}{b}\right).$$

We retain the Gaussian approximation

$$g(x_1) = \varphi(x_1;\, v - \mu,\, \sigma^2),$$

so all closed-form expressions for $c(x_0)$, P^g, and the mixture sampler carry over unchanged from the Gaussian observation case.

9.2. **Tail analysis of $m(0)$.** Writing $t = x_1 - v$, the integrand of $I(0) = \int h(x_1)^2/g(x_1)\, P(x_0, dx_1)$ behaves as

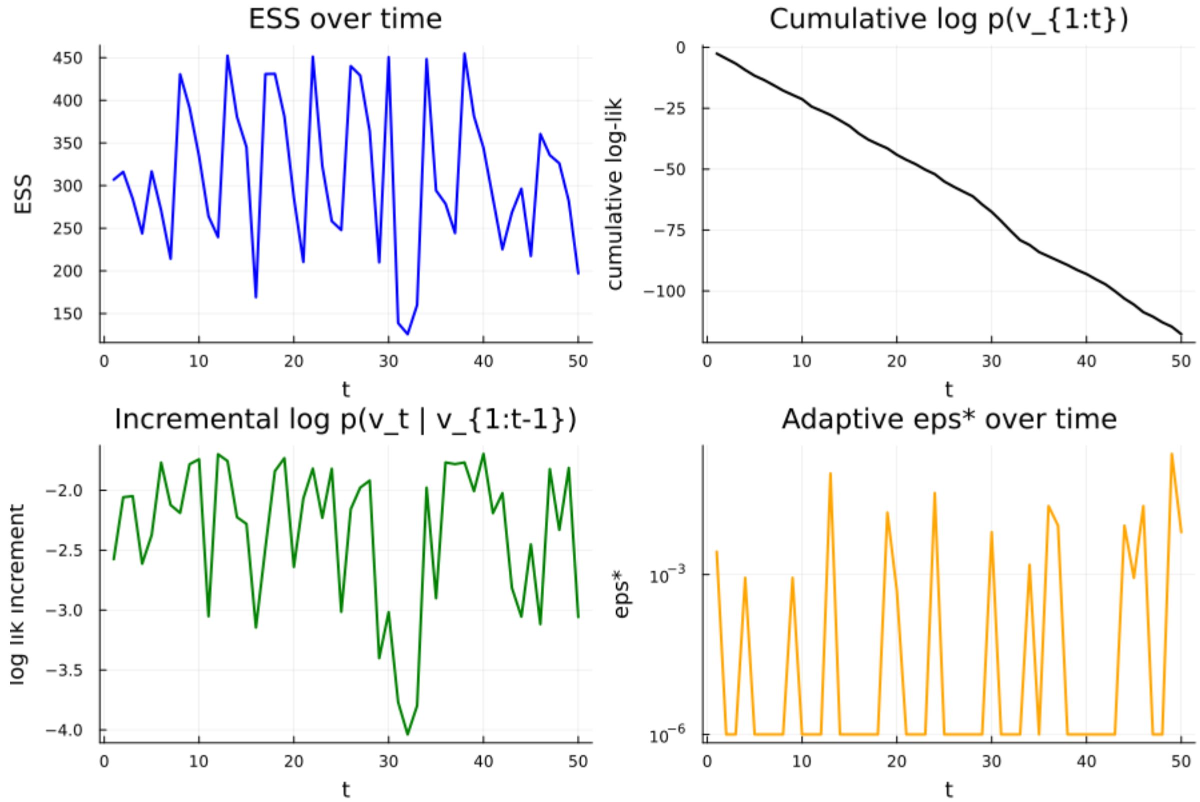
$$\frac{h(x_1)^2}{g(x_1)}\, p(x_1 \mid x_0) \;\propto\; \exp\left(-\frac{2|t|}{b} + \frac{(t+\mu)^2}{2\sigma^2} - \frac{(t - (\omega + \psi x_0 - v))^2}{2\eta^2}\right).$$

- *Right tail $t \to +\infty$.* The linear term $-2t/b$ and the quadratic terms combine to give dominant behaviour

  $$\exp\left(t^2\left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2}\right) - \frac{2t}{b} + O(t)\right).$$

  For this to be integrable we need the coefficient of t^2 to be strictly negative, i.e. $\eta^2 < \sigma^2$, exactly as in the Gaussian observation case. When $\eta^2 \geq \sigma^2$, the quadratic growth dominates the linear decay from the Laplace tails and $I(0) = \infty$.

  When $\eta^2 < \sigma^2$ the quadratic term decays, but the linear term $-2t/b$ provides additional decay that is absent in the Gaussian observation case. The integrand therefore decays faster here, but $I(0)$ is still finite only when $\eta^2 < \sigma^2$.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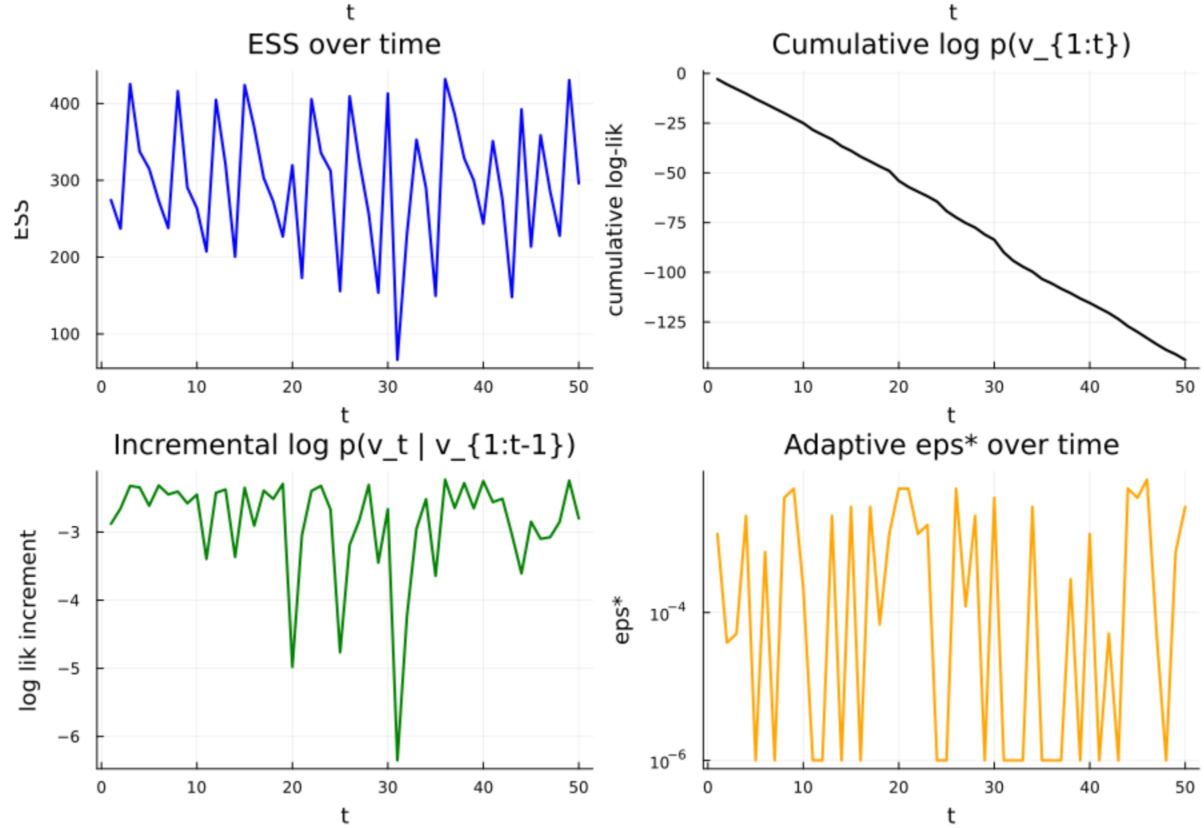
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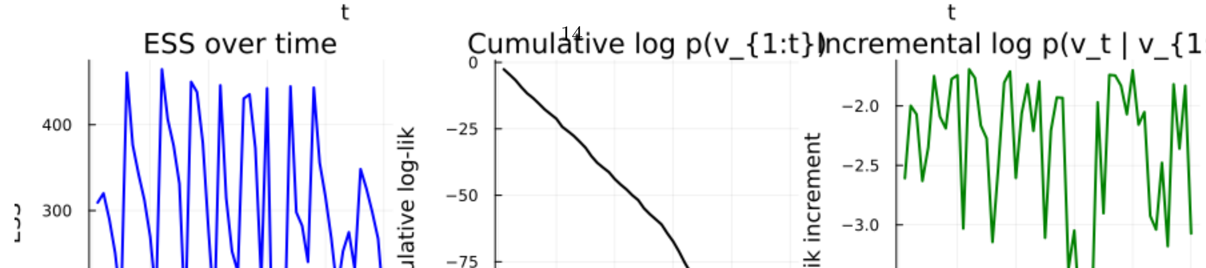
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- *Left tail* $t \to -\infty$. The Laplace density contributes $-2|t|/b = +2t/b \to -\infty$, providing exponential decay. However, when $\eta^2 \geq \sigma^2$ the quadratic term grows, and whether it dominates depends on the sign of $1/(2\sigma^2) - 1/(2\eta^2)$. When $\eta^2 \geq \sigma^2$ the quadratic growth again wins and $I(0) = \infty$.

Proposition 3. For the Laplace observation scheme $V_t = X_t + U_t$, $U_t \sim \text{Laplace}(0, b)$:

$$I(0) = \int \frac{h(x_1)^2}{g(x_1)} P(x_0, dx_1) < \infty \iff \eta^2 < \sigma^2.$$

Consequently, $\bar{m}(0)$ is finite if and only if $\eta < \sigma$, independently of the scale parameter b.

The threshold $\eta = \sigma$ is the same as for the Gaussian observation scheme. However, the Laplace tails make $I(0)$ larger for any fixed $\eta < \sigma$ compared to the Gaussian case, meaning the need for $\varepsilon > 0$ is more pressing even in the finite-variance regime. Furthermore, larger η increases the coefficient of t^2, making $I(\varepsilon)$ larger for any fixed $\varepsilon > 0$ and the optimal ε^* correspondingly larger.

9.3. **The limit $b \downarrow 0$.** As the scale $b \to 0$, the Laplace distribution concentrates on zero and the observation scheme approaches a noiseless observation $V_t = X_t$. In this limit:

- $h(x_1) = \frac{1}{2b} \exp(-|v - x_1|/b) \to \delta_{x_1 = v}$, a point mass at $x_1 = v$.
- The twisted target $P^h(x_0, \cdot)$ concentrates all mass at $x_1 = v$, i.e. the posterior on X_t given $V_t = v$ becomes a point mass.
- The ratio $h(x_1)/g(x_1)$ becomes extremely large near $x_1 = v$ and extremely small away from it, since h is an increasingly sharp spike while g remains spread out. This makes the IS weights degenerate: a single particle near v gets essentially all the weight.
- Consequently, $m(\varepsilon) \to \infty$ for any fixed $\varepsilon \geq 0$ as $b \to 0$, and the optimal $\varepsilon_b^* \to \infty$ as well — no finite regularisation can rescue the IS estimator when the observation is nearly noiseless.

This limiting behaviour highlights a fundamental tension: the more informative the observation (smaller b), the harder the IS problem becomes. Intuitively, the Gaussian approximation g is a poor proxy for a nearly point-mass likelihood, and ε can only partially compensate by mixing in the prior P. In the limit $b = 0$, importance sampling from $P^{g+\varepsilon}$ breaks down entirely and a different approach — such as an exact update using the point-mass likelihood — would be needed.

Numerically, one can verify this by plotting $\bar{m}(\varepsilon)$ for decreasing values of b: the minimum of $\bar{m}$ increases without bound and ε^* drifts toward larger values as $b \to 0$.

10. Pushing the argument to SDEs

Suppose

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + q\,\mathrm{d}W_t, \qquad X_0 = x_0. \tag{2}$$

Here, $b \colon \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}^d$ is a (typically nonlinear) map known as the drift function and $q \in \mathbb{R}^{d \times d'}$ is the diffusion function. Furthermore, W is a standard Brownian motion process in $\mathbb{R}^{d'}$. Assume we observe v at time T and let

$$h(x_T) = \text{density of v conditional on } X_T = x_T.$$

We can

- Make a Gaussian approximation to $h(x_T)$.
- Compute $g(t, x)$ for all t.
- Use guided proposals with not g but $g + \varepsilon$.

Rather than g, take $g_\epsilon := g + \epsilon$. Then

$$\begin{aligned} r_\varepsilon(t,x) := \nabla_x g_\epsilon(t,x) &= \frac{g'(t,x)}{g(t,x)+\epsilon} = \frac{g(t,x)}{g(t,x)+\epsilon} \nabla_x \log g(t,x) \\ &= s(-\log \epsilon + \log g(t,x)) \nabla_x \log g(t,x), \end{aligned}$$

where $s(x) = 1/(1+e^{-x})$. Thus we get a damping factor. The measure $P^{g+\epsilon}$ is the measure (on path space) of the process

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + Q r_\varepsilon(t, X_t) + q\,\mathrm{d}W_t, \qquad X_0 = x_0.$$

with $Q = qq^\top$. Now, with $x = (x_t, \in t \in [0,T])$.

$$G_\varepsilon(x) = h(x_T) \frac{dP}{dP^{g+\varepsilon}}(x)$$

Thus

$$\begin{aligned} m(\varepsilon) = \mathbb{E}_{P^{g+\varepsilon}} G_\varepsilon(X)^2 &= \int G_\varepsilon(x)^2 dP^{g+\varepsilon}(x) \\ &= \int G_\varepsilon(x)^2 \frac{dP^{g+\varepsilon}}{dP}(x) dP(x) \\ &= \int \left(\frac{dP^{g+\varepsilon}}{dP}(x) \right)^{-1} h(x_T)^2 dP(x) \\ &= (g(x_0)+\varepsilon) \int \frac{h(x_T)^2}{g(x_T)+\varepsilon} \Psi(x)^{-1} dP(x), \end{aligned}$$

with

$$\log \Psi(x) = \int_0^T (b(t,x_t) - \tilde{b}(t,x_t)) r_\varepsilon(t,x_t) dt.$$

This is to be compared to (1): c_g in that formula is exactly as $g(x_0)$ here. In the discrete-time we assumed linear dynamics and thus $\tilde{b} = b$ which amounts to $\Psi(x) = 1$. I think the nonlinear case is most interesting here, for else filtering at time T boils down to the discrete-time problem.

Lemma 4. Suppose $g(T,x) = \varphi(v; \mu, \sigma^2)$ (with μ, σ depending on v). If $b(x) = \omega + \psi x$ then for $t \in [0,T)$ we have

$$g(t,x) = ...$$

Closed form expression or via backward ODEs.

Can we establish when $m(0) = \infty$?